National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Linear Algebra	Course Code:	MT1004
Program:	BDS	Semester:	Fall 2021
Duration:	15 Minutes	Total Marks:	10
Paper Date:	December 23, 2021.	Weight	
Section:	1A	Page(s):	
Exam Type:	Quiz-III		

Student : Name:______________________________ **Roll No.**________________ **Section:**_______

Instruction/Notes:

- The maximum time to solve the quiz and upload it to Google Class is 25 minutes.
- Late submissions will not be considered.

Question: Let $\left\{\begin{bmatrix}3\\1\\-1\\3\end{bmatrix},\begin{bmatrix}1\\1\\-2\\8\end{bmatrix},\begin{bmatrix}-5\\1\\5\\-7\end{bmatrix}\right\}$ be a basis for a subspace W of R^4.

a) Use the Gram-Schmidt process to transform this basis to an orthogonal basis for W. Hence find the projection of vector W. **(7)**

b) Obtain the orthogonal complement of subspace W. **(3)**

Solution: